EX. NO: 7

IMPLEMENTATION OF COLUMNAR TRANSPOSITION CIPHER

AIM:

To write a C program to implement the Columnar Transposition Cipher technique.

ALGORITHM:

1. Start the program.
2. Enter the plain text.
3. Enter the key.
4. Remove spaces from the plain text.
5. Write the plain text row by row in a matrix using the length of the key as the number of columns.
6. Number the columns according to the alphabetical order of the key.
7. Read the characters column by column according to the key order.
8. Combine the characters to form the encrypted text.
9. Display the encrypted text.
10. Stop the program.

PROGRAM:

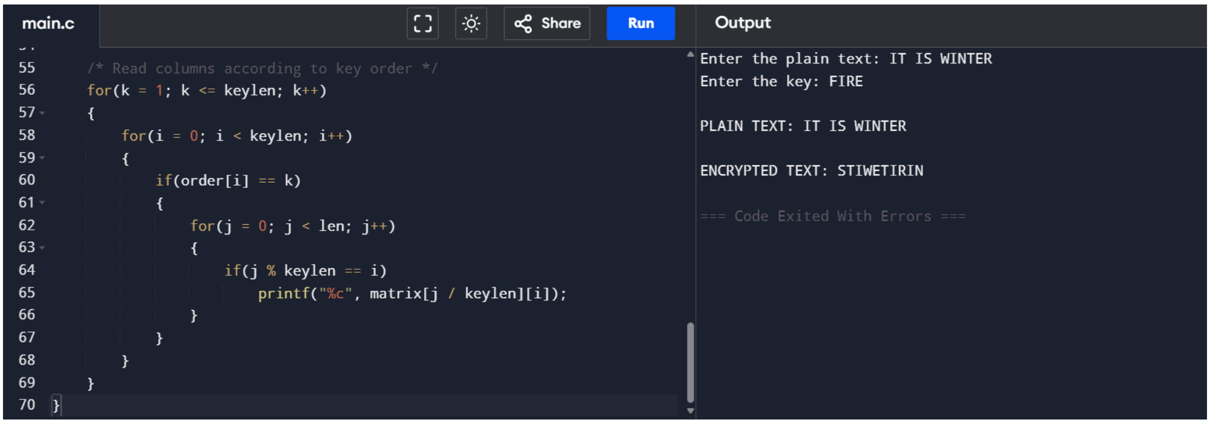
```
#include <stdio.h>

#include <string.h>

#include <ctype.h>


void main()

{

   char plain[100], text[100], key[20];

   char matrix[20][20];

   int i, j, k = 0, len, keylen;
```

```
int order[20];


printf("Enter the plain text: ");
fgets(plain, sizeof(plain), stdin);


/* Remove spaces */
for(i = 0; plain[i] != '\0'; i++)
{
    if(plain[i] != ' ' && plain[i] != '\n')
    {
        text[k++] = plain[i];
    }
}


text[k] = '\0';


printf("Enter the key: ");
scanf("%s", key);


len = strlen(text);
keylen = strlen(key);


/* Find column order */
for(i = 0; i < keylen; i++)
```

```
{
    order[i] = 1;

    for(j = 0; j < keylen; j++)
    {
        if(key[j] < key[i])
            order[i]++;
    }
}

/* Fill the matrix row by row */
k = 0;

for(i = 0; i < len; i++)
{
    matrix[i / keylen][i % keylen] = text[i];
}

printf("\nPLAIN TEXT: %s", plain);
printf("\nENCRYPTED TEXT: ");

/* Read columns according to key order */
for(k = 1; k <= keylen; k++)
{
```

```
        for(i = 0; i < keylen; i++)
        {
            if(order[i] == k)
            {
                for(j = 0; j < len; j++)
                {
                    if(j % keylen == i)
                        printf("%c", matrix[j / keylen][i]);
                }
            }
        }
    }
}
```

OUTPUT:



RESULT:

Thus, the Columnar Transposition Cipher technique was successfully implemented using C language, and the given plaintext was successfully encrypted using the specified key.